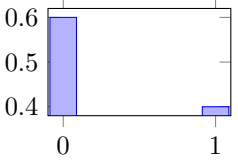
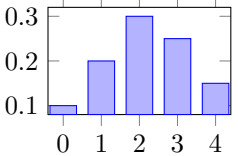
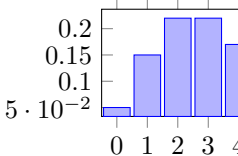
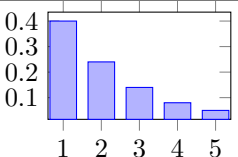
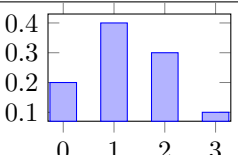
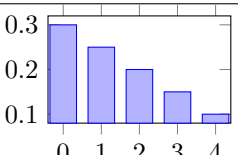
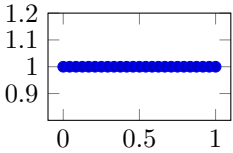
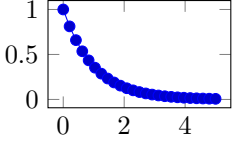
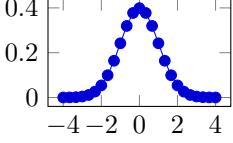
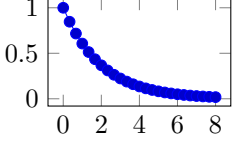
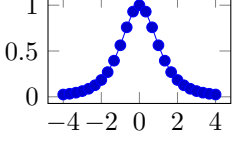
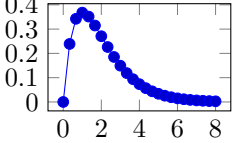


Catálogo de Principales Distribuciones de Probabilidad

Distribuciones Discretas

Distribución	Función de densidad	Esperanza	Varianza	FGM	Gráfica
Bernoulli	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$1-p+pe^t$	
Binomial	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$(1-p+pe^t)^n$	
Poisson	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ	$e^{\lambda(e^t-1)}$	
Geométrica	$(1-p)^{x-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$	
Hipergeométrica	$\frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$n\frac{K}{N}$	$n\frac{K}{N}\frac{N-K}{N}\frac{N-n}{N-1}$	No simple	
Binomial Negativa	$\binom{x+r-1}{r-1}p^r(1-p)^x$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{p}{1-(1-p)e^t}\right)^r$	

Distribuciones Continuas

Distribución	Funcion de densidad	Esperanza	Varianza	FGM	Gráfica
Uniforme	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$	
Exponencial	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$	
Normal	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$	
Chi-cuadrada	$\frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$	k	$2k$	$(1-2t)^{-k/2}$	
t Student	$(1+x^2/\nu)^{-(\nu+1)/2}$	0	$\frac{\nu}{\nu-2}$	No existe	
Gamma	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$(1-\frac{t}{\lambda})^{-\alpha}$	
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	No simple	